What role could decision models play in GeoAI?

Qualifying Spatial Relations using Jev.

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Introduction

Janowicz et al., 2020 asked if there will be an *"artificial GIS analyst"* by 2030. Since then there have been huge strides both in the field of artificial intelligence (AI) and autonomous geoinformation systems (GIS) (Li et al., 2025). But using large language models (LLMs) for spatial analysis requires large amounts of compute, leading to low accessibility and high environmental impact (Janowicz, 2023; Shi et al., 2025). LLMs often produce unnecessary reasoning tokens with high latency and a long response in natural language, which is difficult to process downstream (Chen et al., 2025; Duan et al., 2026; Wang et al., 2026). *Decision models*, also described as *system-one* models, might be more suitable for certain GeoAI (Janowicz et al., 2020) applications, as they operate more efficiently and quickly than LLMs, as well as responding in a predetermined, consistent format. To test this assumption, this thesis reproduces the experiment on *spatial relation qualification* done by Ji et al., 2025 using Typesafe AI's decision model Jev (Almeida, 2026).

Research Goals

The aim of this thesis is to gain initial insights on decision models in GeoAI by reproducing the experiment on spatial relation qualification in Ji et al., 2025. This will enable a comparison between ChatGPT-4 and Jev on the same task. As Jev is not a chat-based model, the prompt engineering methods used in the original experiment, few-shot learning and chain-of-thought reasoning, might not yield the same benefit. Ethical points, especially sustainability, but also bias (Aldahoul and Zaki, 2026), will be discussed too. In total, the thesis links up with works about GeoAI (Janowicz et al., 2020; Ji et al., 2025), autonomous GIS (Li et al., 2025), and ethical use of GeoAI (Janowicz, 2023; Shi et al., 2025). The thesis also wants to contribute to works evaluating and benchmarking Jev (Deußer et al., 2026; Duan et al., 2026; Wang et al., 2026).

The following research questions will be addressed:

- How does the decision model Jev compare with ChatGPT-4 in the task of spatial relation qualification in terms of accuracy and cost?
- In what ways could decision models benefit GeoAI, especially considering sustainability and integration in autonomous GIS?
- Does few-shot prompting and chain-of-thought reasoning improve results for system-one intelligence models?

Methodology

To ensure a good comparison, the thesis will stay as true as possible to the original experiment in Ji et al., 2025, using the same data and setup. The prompts will have to be adapted to the unique requirements of Jev, but the same strategies (few-shot prompting and chain-of-thought reasoning) will be tested, to the extent possible with Jev. The experiment will utilize Typesafe AI's API and Python SDK to access Jev 1.13.0. All code and results will be made available together with the thesis on GitHub. A short review of related works will accompany the experiment along with a discussion on the current state of decision models and their advantages as well as shortcomings for GeoAI.

Expected Results and Challenges

Initial experimentation proved Jev's speed and (cost-)efficiency, while it could not beat the benchmark set by ChatGPT-4 in terms of accuracy. Still, the confidence score, which Jev assigns to its answers, seems reliable, and some interesting patterns occurred in the initial results, which merit further investigation.

The main challenge this thesis will face is a problem common with all LLMs: Their opacity. Jev's answers lack any reasoning and cannot be reproduced reliably. In addition, Jev is a proprietary model, with limited official information released publicly.

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